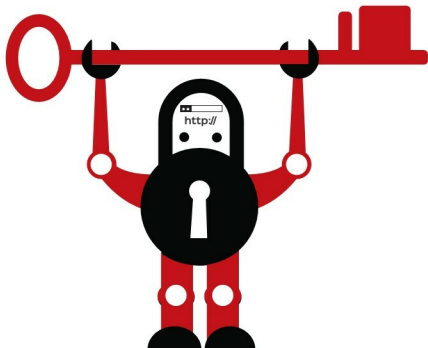


Browser Fingerprinting: How EFF Tools Can Protect You

Even though we think we have secured our browsing by using conventional means, are we really secure from targeted advertising? Is our privacy secured when we browse the World Wide Web? Learn how we can secure our browsing and protect ourselves from browser fingerprinting.



Sign out, clear the cookies, and switch over to the private browsing mode — this is the privacy mantra of most Internet users. If you too fall into this category, then you better be aware that you're being tracked. User accounts and cookies have become archaic when it comes to tracking. You might have noticed personalised ads which are related to your previous search queries, which seem to match your tastes accurately. This happens even when you are not logged in and have no cookies retained. This is made possible by a new tracking technique called *browser fingerprinting*, according to the Electronic Frontier Foundation (EFF).

The EFF is a non-profit organisation founded in 1990, aimed at protecting user privacy and freedom of expression in the digital era. It still fights this battle by conducting campaigns, doing research, and developing security and privacy tools. This is a valuable effort considering the fact that user tracking and immoral use of private information by websites have increased recently. You can learn more about EFF at eff.org.

Thanks to EFF and some software projects, we now have a bunch of free software tools to protect our privacy while browsing the Web. Let's first learn what browser fingerprinting is and find out the EFF tools that present us with the solution.

Note: This article addresses trackers only. Blocking trackers can help you prevent things like personalised ads. But to be truly anonymous, you have to use anonymity networks like Tor, which is a topic beyond the scope of this article. Visit torproject.org for more details, and use it only for legal purposes.

Browser fingerprinting

Browser fingerprinting is a technique that can be employed by websites to track your activities even when you are not signed in and are browsing in private mode. But how does it work?

Well, it's obvious that websites can track and record your activities while you are logged in since they clearly know who you are (at least what your user name is). Even if you are not logged in, they can still drop a cookie in your browser and associate any activity from the same browser with you, which is enough to serve you personalised ads.

That's why we switch over to the private mode, whereby we expect to be immune to the tracking mechanism just because we are not signed in and the browser doesn't retain any cookies from our previous session. The site may still be able to obtain our IP address, but we feel it is of least importance when it comes to identifying us, since the IP